CA/Browser Forum Tracker

Part 4 - Public Trust Programs

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About This Report

This is an experimental, AI-driven research project from SRL. SRL is independent, with no outside funding and no sponsors. It uses several AI models from competing labs, running on different platforms. One model gathers the current claims on a topic, a different one checks each claim against its source, and disagreements are kept, not smoothed over. The aim is an even-handed account that no outside party has shaped. AI makes mistakes, and so does the system built around it. We find and fix some in every cycle, so the methods keep changing and the content should improve over time. Updates come at least quarterly, sooner if something major happens.

The Forum writes the rules, and the root programs decide who is trusted. A root program can and does demand more than the Baseline Requirements, often a year or more ahead of them. This part covers that enforcement layer as it stood on 19 October 2026.

Section 1 covers the four root store policies. Section 2 follows the separate post-quantum trust path that Chrome and Apple are designing. Section 3 covers which authorities are entering and leaving the stores. Section 4 covers the incident record, and Section 5 covers the largest single event of the autumn, GoDaddy’s revocation of 404,607 certificates.

Two kinds of trust removal appear below and should not be confused. One follows a pattern of misconduct and is punitive. The other follows the age of a root and is routine housekeeping. No program took either kind of action between September and 19 October.

1. Root Store Policies

Four organisations run root programs that matter for the public web, and their policies are converging on dedicated hierarchies, bounded root lifetimes and stronger evidence of what an authority actually does.

Recent Developments

Mozilla Root Store Policy version 3.1 was published on 29 June 2026 and took effect on 1 July 2026 [1]. It requires policy and practice documents to be explicit, bounded, auditable and detailed enough to describe issuance [1]. It also requires root key pairs to have been generated within the previous five years [1].

Its largest change arrives later. For audit periods beginning on or after 1 July 2027, authorities with roots enabled for websites must obtain a Detailed Controls Report [1]. On 23 September Mozilla published a whitepaper on what those reports must contain [2]. On the same day it merged a template for mass revocation plans [3]. The template sets thresholds of 100 certificates or 1% of an authority’s active web certificates [3].

The Chrome Root Program Policy is at version 1.8 [4]. It limits the number of roots per owner and requires policy documents to state adherence to the latest policy and to the CCADB Policy from 15 June 2026 [4]. Owners with more than two self-signed roots had to submit a consolidation plan by 15 June 2026, and the cap of two takes effect on 15 September 2027 [4].

Chrome is also removing client authentication from its hierarchies. New subordinate certificates disclosed from 15 June 2026 must carry server authentication only, and new end-entity certificates follow on 15 March 2027 [5]. That timetable comes from a secondary summary of the policy rather than the policy text itself [5].

Microsoft's requirements now cap new roots at ten years of validity [6]. Roots submitted from 1 July 2026 must be dedicated to one purpose, with separate roots for server authentication, email and code signing [6]. Apple's policy version 2.0 took effect on 1 August 2026 [7]. Subordinate certificates signed from that date must carry an extended key usage extension and must not assert the catch-all value [7].

Current State

All four programs are moving against multi-purpose roots. Mozilla requires new roots to serve either websites or email, and existing dual-purpose roots must be split by 31 December 2028 [8]. Chrome requires hierarchies in its store to serve website authentication alone [4].

The policy texts themselves have been quiet since September. The Mozilla, Chrome, Apple and Microsoft policy repositories had no commits between 2 October and 15 October [9]. On 16 October Chrome changed only the testing rules for its quantum-resistant programme, covered in Section 2 [10]. Apple's first change to its repository since August touched only the questionnaire for subordinate authorities run by other companies [11].

Chrome also publishes a statement of direction called Moving Forward, Together [12]. It names automation, term limits for roots, linting, stronger domain validation and shorter lifetimes as goals [12]. It carries no dates of its own, and its items bind only once they reach the policy or a Forum ballot [12].

Unresolved Conflicts

Two version records do not line up. One account attributes Chrome's June 2026 hierarchy rule to version 1.6, while the policy itself is version 1.8 dated 5 February 2026 [4]. Microsoft's requirements file carries a version 1.2 dated 20 May 2026, while its title and earlier records still describe version 1.1 effective 1 July 2026 [6]. Each program's own change history, read in order, would settle which rule entered at which version.

A third conflict concerns Mozilla's mass revocation rule. One source says plans must be tested each year by revoking 30 random certificates within five days, with attestation from April 2026 [13]. Mozilla's template and its earlier guidance date third-party assessment from the June 2025 audit cycle [3] [14]. The current text of the policy's revocation section would settle it.

2. The Post-Quantum Trust Path

A Merkle Tree Certificate replaces the usual chain of signatures with proof that the certificate appears in a signed public log, which keeps large post-quantum signatures out of most connections.

Recent Developments

Chrome and Apple are each designing a separate trust path for post-quantum certificates. Chrome's draft quantum-resistant root program policy, version 0.3.0 from 14 August, describes a root store made up entirely of Merkle Tree Certificate authorities and mirroring operators [15]. Cosigner keys must be ML-DSA-44, held in hardware validated to FIPS 140-3 Level 3 or Common Criteria EAL 4+ [15]. Subscriber certificates may last at most seven days [15].

One clause decides who can issue first. The draft limits initial eligibility to organisations that ran a usable Certificate Transparency log before 1 February 2026 [15]. Chrome opened applications to its testing programme on 31 July, with production planned for 2027 [10]. On 16 October it ruled that new cosigner requests must follow revision 07 of the IETF draft [10]. Revision 05 will keep working for a while, and revision 06 is dropped [10].

Apple announced its own Merkle Tree Certificate policy on 21 September, with a draft due by the end of October [16]. Each certificate would carry three cosignatures, one with ML-DSA and one with ECDSA from the issuer, and a third with ML-DSA from a separate mirror [16]. Validity is capped at seven days, and applications to run such an authority open in late summer 2027 [16]. Apple will also accept composite roots that pair ML-DSA with ECDSA or RSA for email [16].

Microsoft took a different route. On 27 August it deployed seven ML-DSA-87 pilot roots for approved authorities, including DigiCert, Sectigo and SSL.com [17]. These roots are for controlled testing only and are not publicly trusted [17]. They will not be added to the CCADB or to transparency logs [18].

A new issuer then applied. On 29 September Cloudflare said it intends to become a public certificate authority, buying root key material from GlobalSign in a deal it expects to close within two months [19]. It has applied to the Chrome, Apple, Microsoft and Mozilla programs [19]. It aims for inclusion in Chrome's quantum-resistant store by early 2027, a store that does not yet exist in production [19] [10].

Current State

None of these plans is a published rule. Chrome's draft remains at version 0.3.0, and its 16 October change touched only the testing rules [10]. Apple's draft policy had still not appeared on 18 October [11].

The debate has begun anyway. On 29 September a participant on the transparency policy list argued that a seven-day lifetime cap is too large a change to bring in alongside Merkle certificates [16]. In that view it should be announced separately [16].

Chrome has already tied the programme to its security case. After the registries for three country-code domains were hijacked in early October, Chrome named shorter lifetimes, shorter

validation reuse and the quantum-resistant programme as its answers [20]. It gave no new dates [20]. The certificate rules these designs depend on are covered in Parts 1 and 3.

3. Admissions and Removals

Inclusion requests and trust bit removals show who is entering the stores and who is leaving, and each one is argued in public before it happens.

Recent Developments

GoDaddy's two new web roots cleared public discussion on 29 September with no objections [21]. Mozilla posted its intent to approve on 2 October, then deferred approval until seven days after GoDaddy filed an updated practice statement [22]. GoDaddy published version 6.0 on 8 October [23]. On 15 October it confirmed that the certificate profiles in that document did not match what it issues [23]. That admission led to the revocation in Section 5.

eMudhra's four email and web roots reached Mozilla's intent to approve on 2 October, with last call to 9 October [24]. One participant questioned eMudhra's recent incident performance and asked that such notices list a candidate's incidents [24]. A separate bug found seven undisclosed subordinate authorities run by a United Arab Emirates provider under one of eMudhra's roots [25]. Mozilla said it had not yet applied the distrust dates it had announced for two older eMudhra roots [25].

New applicants keep arriving. An inclusion request for two new roots from ISRG, the organisation behind Let's Encrypt, opened on 22 September [26]. IdenTrust opened the public discussion of nine single-purpose roots on 9 October, to run until 20 November [27].

Two authorities asked to leave. Certigna, which stopped issuing public website certificates in June, asked on 13 October for its websites trust bit to be removed [28]. Amazon Trust Services asked the same day for the Extended Validation trust bit to be removed from five of its roots [29].

Current State

No authority was admitted to any root store between September and 19 October. GoDaddy's application is waiting on its incident [30]. eMudhra's application had not changed since 2 October as of 15 October [31]. The IdenTrust discussion has had no posts since 9 October [32].

One standing question is open. In September a participant asked Mozilla what action it considers appropriate over NETLOCK's incident handling [33]. Mozilla replied on 5 October that it is weighing the concerns in its assessment of NETLOCK's continued inclusion, and will follow up publicly [33]. No action has followed.

The precedents are older. In April 2026 Mozilla and Chrome removed DigiCert's three G1 roots for age, not misconduct [34]. The punitive precedent remains the 2024 distrust of Entrust, which Chrome applied by issuance date rather than by revoking existing certificates [35].

4. The Incident Record

Every failure to follow the rules is recorded as a public bug, and the quality of an authority's report is itself part of what the root programs judge.

Recent Developments

September brought several deadlines at once. From 15 September the CCADB Policy required every authority's policy documents to state adherence to it, and enforcement bugs followed within days [36] [37]. The same day, the SHA-1 sunset required the revocation of any unexpired subordinate certificate carrying a SHA-1 signature [38].

Four authorities missed that SHA-1 deadline. GoDaddy revoked two subordinate certificates from 2006 two days late [39]. DigiCert revoked two issuing authorities after a third party reported them, and Firmaprofesional and FNMT each had one left unrevoked [40].

The bug count stayed high through the autumn. Thirty-four compliance bugs opened between 12 and 26 September [40]. Nineteen more followed by 2 October, twelve of them about revocation lists, including bugs against Apple and GoDaddy [41]. Others covered undisclosed issuing authorities at DigiCert and certificates issued to sanctioned entities at Sectigo [41].

October was busier still. Thirty-eight bugs opened between 5 and 15 October [42]. Thirteen concerned failing status services, filed after a third party sent problem reports to many authorities at once [42]. GoDaddy also reported issuing 54 certificates to a key it had already recorded as compromised [43].

Ten more bugs opened on 15 and 16 October, and none on 17 or 18 October [44]. Four were filed together by one member of the public, not by a root program [44]. Each says an audit statement arrived 108 to 144 days late without the required explanation, at ITI Brazil, MULTICERT, Consorci AOC and Entrust [44]. No root program has yet confirmed any of the four [44].

Current State

No program announced a distrust or removal through 19 October [44]. The hijacked country-code registries produced 102 certificates for domains under them, all now blocked through Chrome's revocation list [45]. Chrome said the authorities had not acted improperly [20].

Mozilla's public policy list had no posts between 8 October and 18 October [46]. The incident volume is high, but nothing in it has yet moved a root program to act against an authority.

5. GoDaddy's Mass Revocation

A mass revocation is the moment a root program's rules reach ordinary websites, because every affected certificate must be replaced by its owner.

Recent Developments

On 15 October GoDaddy opened an incident for 404,607 certificates [47]. A third party had found that the policy information in certificates from its GoDaddy and Starfield hierarchies did

not match its certificate policy and practice statement [47]. The Starfield hierarchies hold almost all of them, with 208,896 under one domain validation profile and 187,159 under another [47]. GoDaddy stopped issuance, corrected the profiles and resumed issuance on the corrected ones [47].

The revocation deadline was 18 October at 23:45 UTC [48]. On 16 October GoDaddy said it did not expect to meet it, and filed the delay as a separate incident [48]. It said it had begun contacting subscribers on 14 October [48].

Current State

GoDaddy missed the deadline. Its 19 October update reported 170,566 certificates revoked by 23:45 UTC on 18 October [48]. By 05:29 UTC on 19 October the total was 227,422 of 404,607 [48]. About 177,000 remained.

The update gave no expected completion date [48]. The next update is due on Thursday 22 October 2026 [48]. GoDaddy's root inclusion application remains open while the incident runs [30].

Unresolved Conflicts

How long the delay runs, and how Mozilla weighs it against the pending inclusion, are both open. The 22 October update and any root program comment on the delayed revocation bug would settle the first. A decision posted on the inclusion request would settle the second.

Sources

1. Improving Transparency and Assurance in the Web PKI, Mozilla Root Store Policy v3.1 - Mozilla Security Blog - https://blog.mozilla.org/security/2026/06/29/improving-transparency-and-assurance-in-the-web-pki-mozilla-root-store-policy-v3-1/ - as of 2026-07-01
2. Pull request 303, Detailed Controls Reports whitepaper Version 1.0 - mozilla/pkipolicy - https://github.com/mozilla/pkipolicy/pull/303 - as of 2026-09-23
3. Mass Revocation Template - mozilla/pkipolicy - https://raw.githubusercontent.com/mozilla/pkipolicy/master/rootstore/Mass-Revocation-Template.md - as of 2026-09-23
4. Chrome Root Program Policy, Version 1.8 - Google Chrome - https://googlechrome.github.io/chromerootprogram/crp/policy - as of 2026-09-27
5. Chrome Root Program Policy v1.8: Full 2026-2027 Timeline - Encryption Consulting - https://www.encryptionconsulting.com/chromes-root-program-policy-v1-8/ - as of 2026-09-12
6. Microsoft Root Program Requirements - TrustedRootProgram - https://github.com/TrustedRootProgram/Program-Requirements/blob/main/Requirements.md - as of 2026-09-27
7. Apple Root Program Policy - Apple - https://github.com/apple/apple-root-program/blob/main/policy.md - as of 2026-09-27

8. Mozilla Root Store Policy - Mozilla - https://www.mozilla.org/en-US/about/governance/policies/security-group/certs/policy/ - as of 2026-07-01
9. Commits, mozilla/pkipolicy - GitHub - https://api.github.com/repos/mozilla/pkipolicy/commits - as of 2026-10-15
10. Chrome Root Program commit 5f8d529, quantum-resistant root program testing FAQ - GitHub - https://api.github.com/repos/GoogleChrome/chromerootprogram/commits/5f8d529 - as of 2026-10-16
11. Apple Root Program policy repository tree - GitHub - https://api.github.com/repos/apple/apple-root-program/git/trees/main?recursive=1 - as of 2026-10-18
12. Moving Forward, Together - Chrome Root Program - https://googlechrome.github.io/chromerootprogram/moving-forward-together/ - as of 2026-09-01
13. Add section 6.1.3 for Delayed Revocation - mozilla/pkipolicy - https://github.com/mozilla/pkipolicy/commit/efa8ac40ac341fb813620938ef72328a53858038 - as of 2026-09-12
14. Responding to a misissuance, mass revocation requirements - MozillaWiki - https://wiki.mozilla.org/CA/Responding_To_A_Misissuance - as of 2025-09-01
15. [DRAFT] Chrome Quantum-resistant Root Program Policy, Version 0.3.0 - Google Chrome - https://googlechrome.github.io/chromerootprogram/cqrp/draft-policy - as of 2026-08-14
16. Apple Root Program PQC Announcement - chromium ct-policy list - https://groups.google.com/a/chromium.org/g/ct-policy/c/QGZw2ADMXvk - as of 2026-09-29
17. Microsoft Trusted Root Program announcements, PQC TLS pilot roots - TrustedRootProgram - https://github.com/TrustedRootProgram/Program-Requirements/blob/main/Announcements.md - as of 2026-08-27
18. Post-Quantum Cryptography (PQC) TLS Pilot Program Requirements V1.0 - TrustedRootProgram - https://github.com/TrustedRootProgram/Program-Requirements/blob/main/PQC%20Pilot%20Program.md - as of 2026-09-26
19. Building a post-quantum certificate authority with Merkle Tree Certificates - Cloudflare Blog - https://blog.cloudflare.com/pq-ca-with-mtcs/ - as of 2026-09-29
20. Chrome’s response to recent ccTLD registry hijacks - Google Security Blog - https://blog.google/security/chromes-response-to-recent-cctld-registry-hijacks/ - as of 2026-10-06
21. Public discussion of GoDaddy root inclusion request - CCADB public list - https://groups.google.com/a/ccadb.org/g/public/c/6hPxKjObVlc - as of 2026-09-29
22. Notice of Intent to Approve, GoDaddy root inclusion - mozilla dev-security-policy - https://groups.google.com/a/mozilla.org/g/dev-security-policy/c/XlUPVy2F0gs - as of 2026-10-02
23. Bug 2035858, GoDaddy root inclusion request - Mozilla Bugzilla - https://bugzilla.mozilla.org/show_bug.cgi?id=2035858 - as of 2026-10-15

24. Notice of Intent to Approve, eMudhra root inclusion - mozilla dev-security-policy - https://groups.google.com/a/mozilla.org/g/dev-security-policy/c/h2ucXHS5HZ4 - as of 2026-10-02
25. Bug 2018979, eMudhra root transition - Mozilla Bugzilla - https://bugzilla.mozilla.org/show_bug.cgi?id=2018979 - as of 2026-10-13
26. CA Certificate Root Program bugs changed from 25 September 2026 - Mozilla Bugzilla - https://bugzilla.mozilla.org/rest/bug?product=CA%20Program&component=CA%20Certificate%20Root%20Program&last_change_time=2026-09-25&include_fields=id,summary,creation_time,last_change_time,status,resolution - as of 2026-10-04
27. Public discussion of IdenTrust root inclusion request - CCADB public list - https://groups.google.com/a/ccadb.org/g/public/c/eyFzjLZdAKI - as of 2026-10-09
28. Bug 2081248, Certigna trust bit removal request - Mozilla Bugzilla - https://bugzilla.mozilla.org/show_bug.cgi?id=2081248 - as of 2026-10-13
29. Bug 2081415, Amazon Trust Services EV trust bit removal request - Mozilla Bugzilla - https://bugzilla.mozilla.org/show_bug.cgi?id=2081415 - as of 2026-10-13
30. Bug 2035858, GoDaddy root inclusion request - Mozilla Bugzilla - https://bugzilla.mozilla.org/show_bug.cgi?id=2035858 - as of 2026-10-18
31. Bug 1889859, eMudhra root inclusion request - Mozilla Bugzilla - https://bugzilla.mozilla.org/show_bug.cgi?id=1889859 - as of 2026-10-15
32. CCADB public list - CCADB - https://groups.google.com/a/ccadb.org/g/public - as of 2026-10-18
33. Netlock inclusion in MRSP - mozilla dev-security-policy - https://groups.google.com/a/mozilla.org/g/dev-security-policy/c/a7d2tppkX-A - as of 2026-10-05
34. DigiCert G1 root removal advisory - DigiCert - https://docs.digicert.com/en/certcentral/order-and-manage-certificates/reports-and-advisories/review-the-g1-root-removal-advisory.html - as of 2026-04-15
35. Sustaining Digital Certificate Security, Entrust Certificate Distrust - Google Online Security Blog - https://security.googleblog.com/2024/06/sustaining-digital-certificate-security.html - as of 2024-11-12
36. CCADB Policy, Version 2.1 - CCADB - https://www.ccadb.org/policy - as of 2026-09-15
37. Bug 2073388, SHECA CP/CPS missing root program adherence statements - Mozilla Bugzilla - https://bugzilla.mozilla.org/show_bug.cgi?id=2073388 - as of 2026-09-23
38. Ballot SC097, Sunset all remaining use of SHA-1 signatures in certificates and CRLs - CA/Browser Forum - https://cabforum.org/2026/01/24/ballot-sc097-sunset-all-remaining-use-of-sha-1-signatures-in-certificates-and-crls/ - as of 2026-09-15
39. Bug 2072663, GoDaddy delayed revocation of SHA-1 signed subordinate CA certificates - Mozilla Bugzilla - https://bugzilla.mozilla.org/show_bug.cgi?id=2072663 - as of 2026-09-18
40. CA Certificate Compliance bugs opened from 12 September 2026 - Mozilla Bugzilla - https://bugzilla.mozilla.org/rest/bug?

product=CA%20Program&component=CA%20Certificate%20Compliance&creation_time=2026-09-12 - as of 2026-09-26

41. CA Certificate Compliance bugs opened from 27 September 2026 - Mozilla Bugzilla - https://bugzilla.mozilla.org/rest/bug?product=CA%20Program&component=CA%20Certificate%20Compliance&creation_time=2026-09-27&include_fields=id,summary,creation_time,status - as of 2026-10-02
42. CA Certificate Compliance bugs opened from 3 October 2026 - Mozilla Bugzilla - https://bugzilla.mozilla.org/rest/bug?product=CA%20Program&component=CA%20Certificate%20Compliance&creation_time=2026-10-03&include_fields=id,summary,creation_time,status,whiteboard - as of 2026-10-15
43. Bug 2079943, GoDaddy certificates issued to a compromised key - Mozilla Bugzilla - https://bugzilla.mozilla.org/show_bug.cgi?id=2079943 - as of 2026-10-15
44. CA Certificate Compliance bugs opened from 15 October 2026 - Mozilla Bugzilla - https://bugzilla.mozilla.org/rest/bug?product=CA%20Program&component=CA%20Certificate%20Compliance&creation_time=2026-10-15&include_fields=id,summary,creation_time,status,whiteboard - as of 2026-10-18
45. ccTLD Registrar Hijacks - mozilla dev-security-policy - https://groups.google.com/a/mozilla.org/g/dev-security-policy - as of 2026-10-08
46. dev-security-policy list - Mozilla - https://groups.google.com/a/mozilla.org/g/dev-security-policy - as of 2026-10-18
47. Bug 2082703, GoDaddy certificate policy information mismatch - Mozilla Bugzilla - https://bugzilla.mozilla.org/show_bug.cgi?id=2082703 - as of 2026-10-19
48. Bug 2083191, GoDaddy delayed revocation - Mozilla Bugzilla - https://bugzilla.mozilla.org/show_bug.cgi?id=2083191 - as of 2026-10-19